Commercial vs. Government Rocket Launch Reliability

Mid-semester check-in

Team Rocket

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STA 9750, Fall 2026



The overarching question

From 2000–2025, how does the reliability of launches differ between commercial and government providers, and what factors drive and help explain the difference?

Our starting measure

Successful launches
÷ (success + failure + partial failure)

Partial failures are kept as their own outcome.

What we look at

- **SQ1** Mission context
- **SQ2** Change over time
- **SQ3** Launch experience
- **SQ4** Failure concentration
- **SQ5** Recovery after failure

Two public sources with highlights and limitations

Launch Library 2: Main Source

The Space Devs' free launch API (2.3.0): 5 endpoints and 7 tables

Contains

- 2,960 launches and 66 columns, 2000-2025
- One row per launch with date, outcome, provider and sector, rocket, mission, orbit, launch site

Fit

- Good and direct fit for our OQ and SQs: sector, outcome and rocket for every launch
- Three outcome levels: success, partial failure, failure

Limitations

- Recording: 11% missing orbit/mission type, 70 of 164 failures undescribed, 3 errors (found with GCAT)
- Sampling: only popular suborbital launches; misses 2 orbital government tests
- Classification: sector = provider's current type

GCAT: Secondary

J. McDowell's catalog (planet4589.org): free TSV, Oct. 24, 2026 snapshot

Contains

- 6,793 launches and 33 columns, 2000-2025 (76,014 since 1942)
- Official launch tag, date, rocket type, launch code (flight type, outcome, success score), failure code, service code

Fit

- Independent catalog: matches 2,801 of 2,802 orbital launches (cross-checks LL2)
- 99.7% outcome agreement with LL2
- Failure code for all 137 GCAT failures in the data
- Success score for 36 launches (13 of 23 LL2 partial failures)

Limitations

- Recording: some failure codes are unknown (U, UX, S3U)
- Definitions: scores the rocket, not the mission (mission in LL2) and service code is not the same concept as our sector
- Sampling: matches 71% of LL2's suborbital launches

Note: sometimes the rocket reaches orbit but doesn't fully complete the mission. One source may label that a partial failure, while the other counts it as a success with qualifications.

How we joined LL2 and GCAT into one master file

Launch Library 2
Main dataset

GCAT
J. McDowell's catalog

How we joined them

1 Exact ID match

LL2 launch_designator = GCAT Launch_Tag

2,690 matched

2 Same-day match

No ID – one GCAT launch that day

192 matched

3 Manual review

Checked by rocket and date

31 matched

Total = 2,913 matched
(2,801 orbital + 112 suborbital)

Joined master file

2,959 launches and 77 columns
LL2 kept, GCAT added, 1 duplicate removed (counted twice in LL2)

2,801 of 2,802
Orbital launches matched (exception: Amos-6, pad explosion)

What GCAT adds

- **How launches failed**
  Failure code for all 137 GCAT failures (a few unknown: U, UX, S3U)
- **A second view of the sector**
  A check on our commercial / government labels
- **Error checks on LL2**
  Caught a duplicate, a wrong ID and a wrong date

Limits: completeness, a strong match within orbital launches does not establish complete coverage of all launches - neither source is guaranteed complete

Overall difference

SQ 1 • Nevena Kolarevic

BEFORE

How do commercial and government providers compare on success, partial-failure, and complete failure rates?

AFTER

How do commercial and government providers compare on success, partial-failure, and complete-failure rates for similar launches, and how do their failures differ in type and severity?

	Why we changed it	How
Overall Rate	One rate per sector can hide real differences	Rates by orbit, mission type and rocket size
Mission Mix	The sectors don't fly the same types of launches	Compare similar launches and shared rockets
Failures	Success vs failure doesn't show what went wrong	GCAT failure codes, where available
Partial Failures	Feedback warned us not to oversimplify them	Kept as their own outcome throughout

Note: about 11% of launches are missing orbit or mission type, and GCAT failure codes don't cover every failure, so some groups will be small

Change over time

SQ 2 • Nastassjia Durand

BEFORE

How has the commercial-government reliability gap changed over time?

AFTER

How has the reliability gap between commercial and government launches changed from 2000 to 2025, and are the shifts associated with specific countries or with new providers entering the market?

	Why we changed it	How
Scope	Feedback: the original question was too narrow	Added countries and new providers
Over Time	One 25-year average hides shifts	Yearly success rate by sector + 3-year rolling rate
Countries	Each country's launch mix changed over 25 years	Rolling rate by sector and country
New Providers	Many commercial entrants since the 2010s; early flights fail more often	New vs established providers

Note: new vs. established providers will be evaluated based on N previous launches, where fewer than N = new (N to be tested); SpaceX alone makes up 39% of commercial launches

Launch experience

SQ 3 • Kelly Kim

BEFORE

How does reliability change as a rocket configuration gains launch experience, and does that pattern differ by provider type?

AFTER

How do launch outcomes and failure mechanisms change from early to later configuration flights, and do these patterns differ between commercial and government providers?

	Why we changed it	How
Feedback	Go beyond success vs failure (stages, partial failures)	Track success, partial and failure, plus GCAT failure codes
Fair Comparison	Configurations launch at very different volumes	Compare at similar levels of prior launch experience
Experience	Early and mature configurations may have different reliability patterns	Group launches into early and later configuration flights within 2000-2025
Provider Type	Commercial and government may learn differently	Compare experience-outcome patterns by sector

Note: experience = launches of the same configuration, not reuse of one booster

Failure concentration

SQ 4 • Andrew Wilber

BEFORE

Are failures concentrated among a small number of rocket configurations, or distributed across many vehicles?

AFTER

Which configurations have demonstrated disproportionately high failure rates, and to what extent can development stage and recurring technical problems explain those patterns?

	Why we changed it	How
Launch Volume	Some rockets launch more often than others	Compare each configuration's share of launches and failures
Early Testing	Problems during initial testing may not continue over time	Examine failures throughout each configuration's life cycle
Technical Causes	Problems during initial testing may not continue over time	Use GCAT failure codes an investigation reports

Note: some configurations have relatively few launches and the available data does not always explain what caused the failure

Recovery after failure

SQ 5 • Christopher Orellana

BEFORE

After a failure, how many launches or how much time does a provider typically need to return to a successful launch?

AFTER

How does the time and number of launches needed to return to success differ between commercial and government providers, and how does recovery vary by failure type and the rocket configuration involved?

	Why we changed it	How
Sector Comparison	The original asked about typical recovery, without directly comparing sector specifics	To examine how recovery varies by provider type, failure type, and rocket configuration
Failure Detail	The original treated all failures alike	Compare recovery by GCAT failure type
Rocket Design	A provider's next success may involve a different rocket	Compare provider and same-configuration recovery

Prior work and our contribution

What earlier work covers	The gap	What we propose to add
Wagenblast & Bettinger (2024) - Launch reliability averaged ~95% annually - Generally higher for vehicle families with more launch experience	Paid data, ends 2023 and no focus on how the sector gap changed	**Updated, reproducible evidence** - How reliability gap by sector changed over time - Analyzes launches from 2000-2025 - Uses LL2 & GCAT (3 outcome levels: success, partial failure and complete failure + failure codes)
Moore, Aerospace (2019) - Launch reliability generally increased with vehicle-family experience - Medium-capacity vehicles historically performed better than small and heavy-capacity vehicles	U.S. & Europe only and no commercial vs. government comparison	**Experience by sector** - Compares outcomes in early vs. later configuration flights across commercial and government providers
Go et al., NASA (2017) - U.S. launch failures 1980-2015 have different causes and consequences, and differ by rocket stage and propulsion type	No link to recovery or to sector and U.S. only, ends 2015	**Failure types & recovery** - Measures days and number of launches until the next success - Includes failures with no observed recovery by 2025 - Compares failure types

Deeper outcomes and remaining challenges

Beyond the overall success rate

Retain partial failures

Separate payload delivery from booster landing outcomes

Use GCAT codes where available

What still needs care

Different outcome definitions	Compare LL2 and GCAT, review disagreements
Unknown failure codes and failure coverage	Keep them as unknown
Orbital, suborbital and test-flight scope	Team decision to do
Unequal launch volume	Compare failure rates, not just counts
No observed recovery by the cutoff	Count launches by providers, not per rocket
Sector labels (76 unknown)	Manual fix (cross-check with GCAT service codes)

Next: **verify the enriched master joined file, agree on classifications, and finalize each revised SQ**

References

Launch Library 2

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